PAPER<i>351 ■ ASASSN-14li Tidal Disruption Event: Ultrafast Outflow F</i>U<i>Bi</i>i and Kozima LENR Force

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Session: *96*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF F*U*Bi_i for a TDE with 0.3c ultrafast outflow and Kozima LENR coupling **Author:** Daniel T. Murphy

<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

ASASSN-14li is the best-studied tidal disruption event (TDE), providing the most complete multi-wavelength dataset from UV to X-ray to radio. The UQFF buoyancy-unified force is computed for the stellar mass black hole remnant (M*BH = 106 M?), yielding F*U*Bi*i ■ -8.32■10■■■ N ■ six orders of magnitude smaller than AGN-scale F*U*Bi*i, reflecting the much smaller BH mass. The ultrafast outflow at* vout = 0.3c is connected to UQFF via the Kozima LENR force component F_Kozima = 10■■ N at the stellar disruption interface.

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2. Core Physics

2.1 UQFF Buoyancy-Unified Force (TDE Scale)

```
F_{U\_Bi\_i} \approx -8.32 \times 10^{211}\ \mathrm{N}
```

The six-order-of-magnitude reduction from the AGN scale (-8.32■10■■7 N) reflects M_BH = 106 M? vs. 10? M?.

2.2 Ultrafast Outflow

```
v_{\rm out} = 0.3c = 9.0 \times 10^7\ \mathrm{m/s}
```

Observed in Chandra/XMM-Newton blueshifted Fe K absorption lines. The UQFF kinetic coupling:

```
P_{\rm outflow} = \frac{1}{2} \dot{M}_{\rm out} v_{\rm out}^2 = \frac{1}{2} \dot{M}_{\rm out} (0.3c)^2
```

2.3 Kozima LENR Force Component

```
F_{\rm Kozima} = 1 \times 10^{30}\ \mathrm{N}
```

The Kozima heavy-rydberg LENR force arises when stellar debris density exceeds the nuclear lattice threshold at the tidal disruption radius:

```
r_{\rm tide} = R_\star \left(\frac{M_{\rm BH}}{M_\star}\right)^{1/3}
```

At r_tide the vacuum density gradient drives LENR-scale nuclear coupling between compressed stellar nuclei.

2.4 Full FUBi_i Decomposition

```
F_{U\_Bi\_i}^{\rm ASASSN} = F_{\rm UQFF}^{\rm TDE} + F_{\rm Kozima} + F_{\rm outflow}
```

```
\approx -8.32\times 10^{211} + 10^{30} + P_{\rm outflow}/r_{\rm tide}\ \mathrm{N}
```

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3. Key Values

Quantity	Formula	Value
M_BH	UV-optical fit	106 M?
F*U*Bi_i	UQFF TDE scale	-8.32■10■■■ N
v_out	Chandra Fe K	0.3c
F_Kozima	LENR coupling	10■■ N
r_tide	R*?(M*BH/M_?)^(1/3)	~7 R?

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4. Physical Significance

ASASSN-14li bridges stellar-scale and AGN-scale UQFF physics. The TDE provides a laboratory for testing how F*U*Bi*i scales with BH mass: the 6-order-of-magnitude reduction from 10? M? to 106 M? tracks the mass scaling F*UBi*i ? M*BH^a, a derived from comparing PAPER*346 (M87) to PAPER_351, enabling a power-law calibration of the BH mass dependence of UQFF vacuum buoyancy.

The Kozima LENR force at F*Kozima = 10■■ N is much smaller than F*UBi*i in this TDE context, suggesting LENR effects are perturbative at stellar BH scales.

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5. Deduplication Note

vs. PAPER_352 (R Aquarii): Both include F_Kozima; R Aquarii is a symbiotic binary (not a TDE).

vs. all AGN papers (346■350): TDE F*U*Bi_i ■ 10■■■ N (stellar mass BH) vs. AGN 10■■7■10■■8 N.

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6. Classification

Physics Territory: FIRST UQFF TDE with ultrafast outflow (0.3c) and Kozima LENR coupling **Scale:** Stellar (106 M? BH ■ TDE disruption radius) **CP Implementation:** `ASASSN14liTDEOutflowFUBiCalculator` (CondensedPhysics3.py, Session 96)

Testable Prediction: This UQFF result is directly testable with NICER/Chandra (X-ray; testable at 3s by 2027); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.
